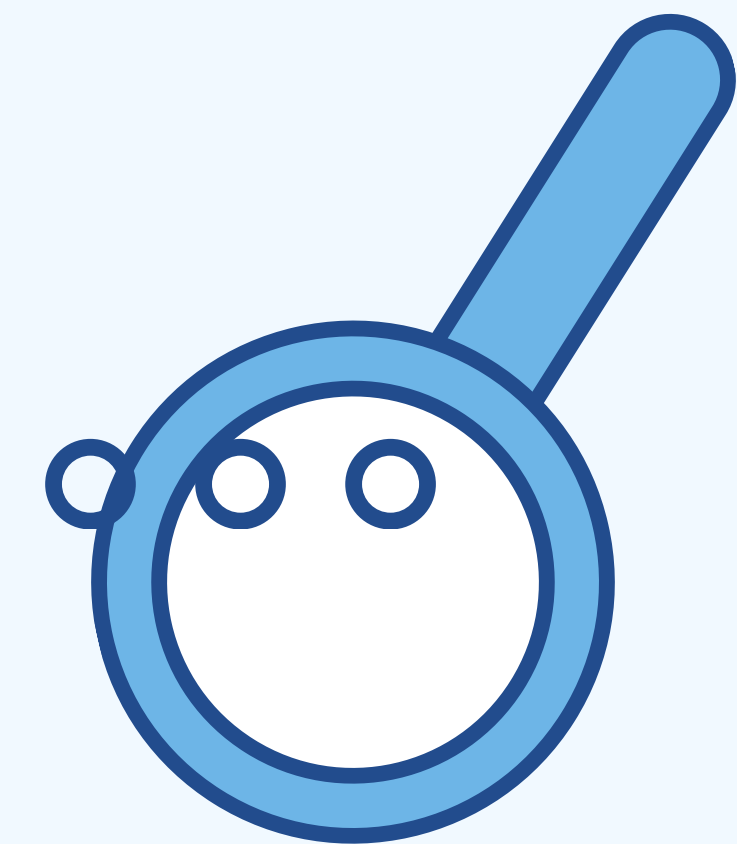
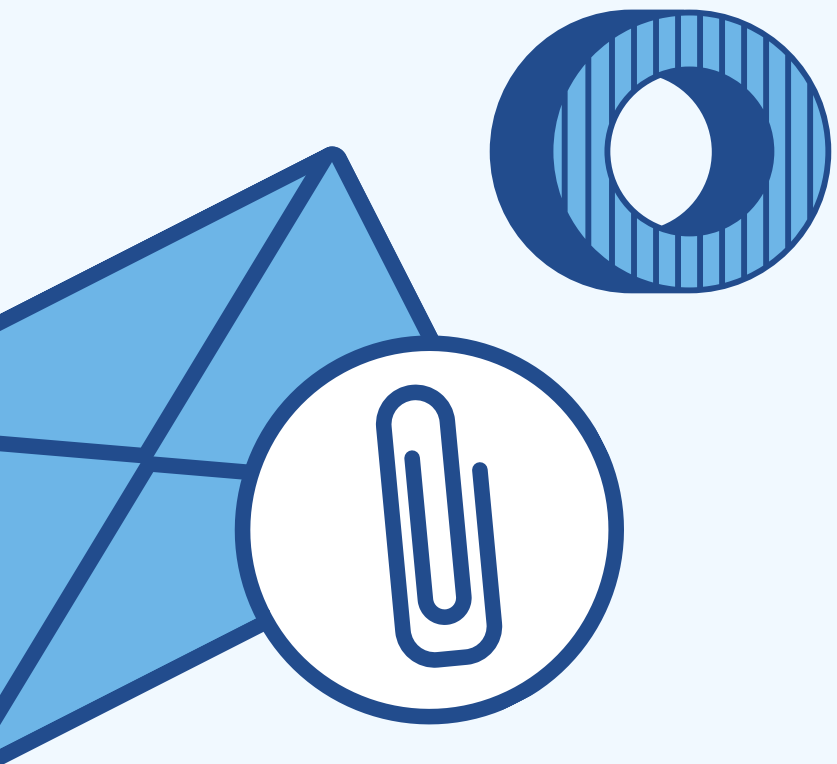




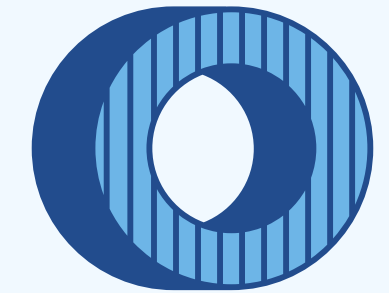
O + PUZZLEBOT (POSE V. ANGULARES)



EZEQUIEL LUNA
OMAR PEREZ
ANTONIO MENDEZ



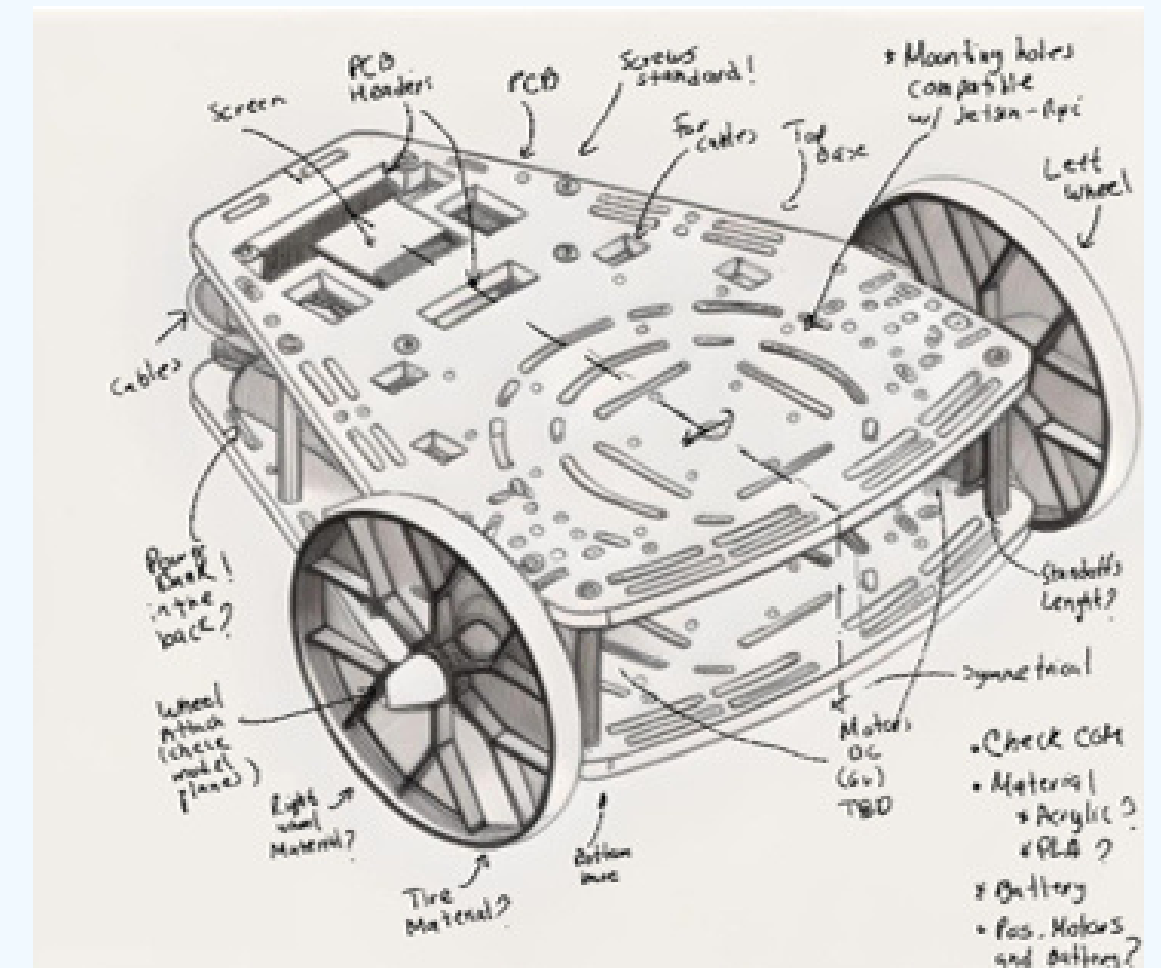
Puzzlebot (especificaciones)



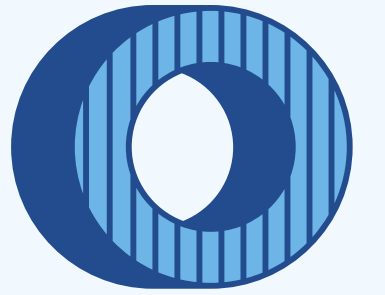
Robot de traccion diferencial

Parametros fisicos

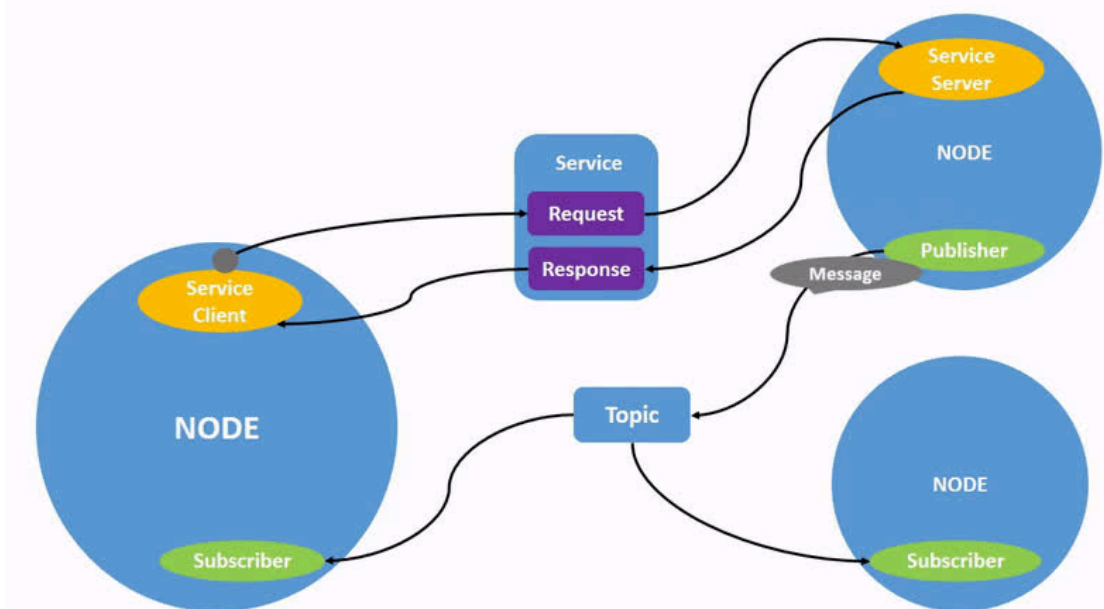
Radio de ruedas: 5cm y Distacia entre ejes
17 cm



Topico cmd/vel



- Canal universal para el control de movimiento base.
- Mensaje: geometry_msgs/msg/Twist.
- Componentes Activos (Puzzlebot):
Velocidad Lineal en X Avance/Retroceso.
Velocidad Angular en Z Giros sobre su propio eje.



Modelo cinematico inverso



- r: radio de la rueda
- L: distancia entre ruedas
- ω_R , ω_L : velocidades angulares de cada rueda
- v: velocidad lineal del robot
- ω : velocidad angular del robot

1.-Velocidad lineal es el promedio de las velocidades lineales de cada llanta

$$v = \frac{Vr + Vl}{2}$$

$$v = r * w$$

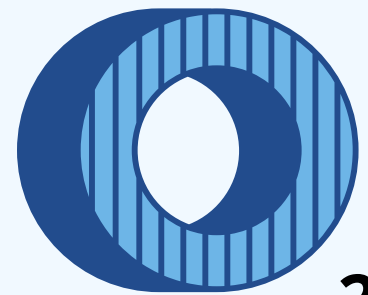
$$v = \frac{r * \omega_R + r * \omega_L}{2}$$



2.-La rotación depende de la diferencia entre las ruedas

$$\omega = \frac{\omega_R r - \omega_L r}{L}$$

$$\omega = \frac{r * \omega_R - r * \omega_L}{L}$$



Modelo cinematico inverso



3.-Sistema de ecuaciones

4.-Despejamos y multiplicamos

$$v = \frac{r*wr + r*wl}{2}$$

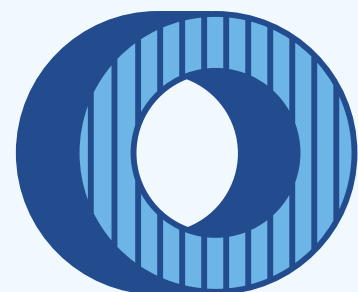
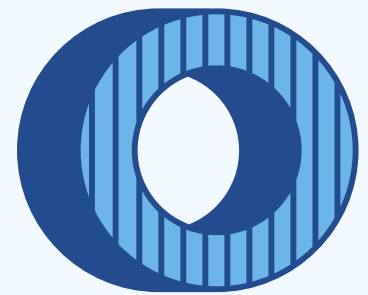
$$w = \frac{r*wr - r*wl}{L}$$

$$2v = r(wr + wl)$$

$$wL = r(wr - wl)$$

$$2v = rwr + rwl$$

$$wL = rwr - rwl$$



Modelo cinematico inverso



5.-Sumando ambas ecuaciones
obtenemos w_r

$$2v + wL = (rwr + rwl) + (rwr - rwl)$$

$$2v + wL = 2r * w_r$$

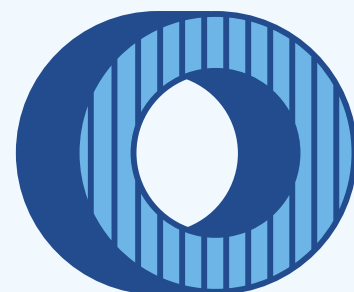
$$w_r = \frac{v}{r} + \frac{L}{2r}w$$

6.- Restando ambas ecuaciones
obtenemos w_l

$$2v - wL = (rwr + rwl) - (rwr - rwl)$$

$$2v - wL = 2r * w_l$$

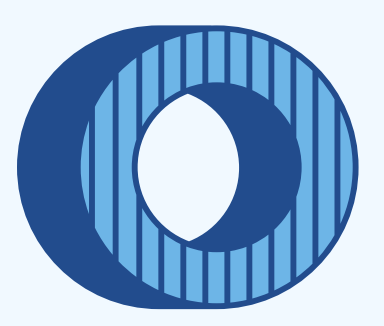
$$w_l = \frac{v}{r} - \frac{L}{2r}w$$



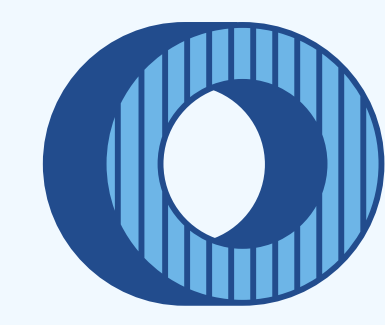
Modelo cinematico directo



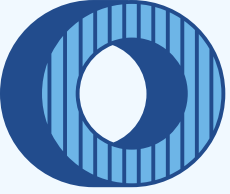
1.-Ecuaciones diferenciales


$$\begin{aligned}x' &= v * \cos(phi) \\ y' &= v * \sin(phi) \\ phi &= w\end{aligned}$$

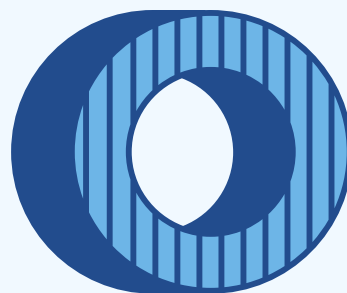
2.-metodo de integracion de euler


$$\begin{aligned}phi_k + 1 &= phi_k + w_k * t_s \\ x_k + 1 &= x_k + (v_k * \cos(phi_k)) * t_s \\ y_k + 1 &= y_k + (v_k * \sin(phi_k)) * t_s\end{aligned}$$

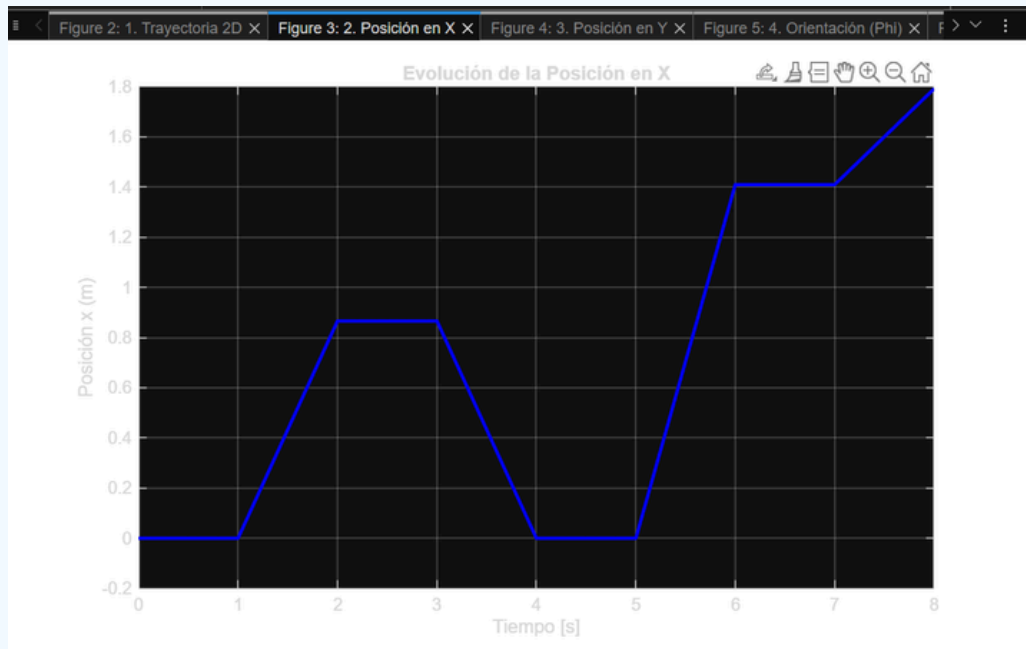


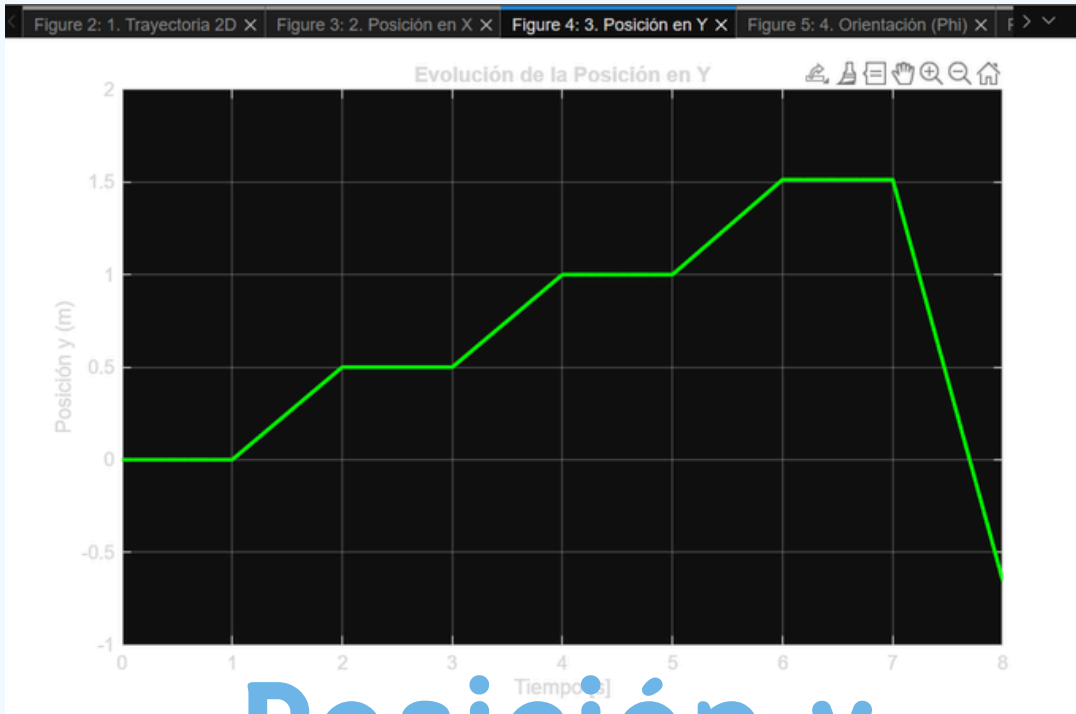
Analisis de graficas



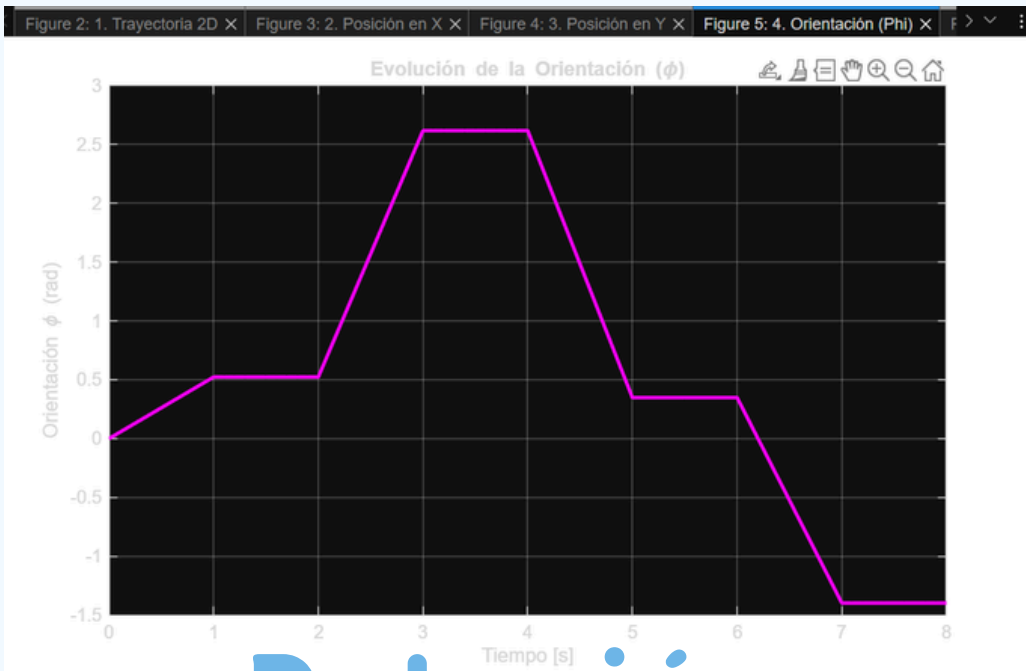
Trayectoria



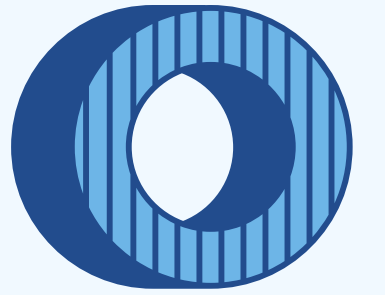
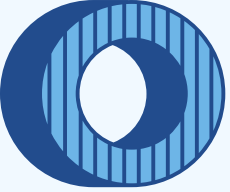
Posición X



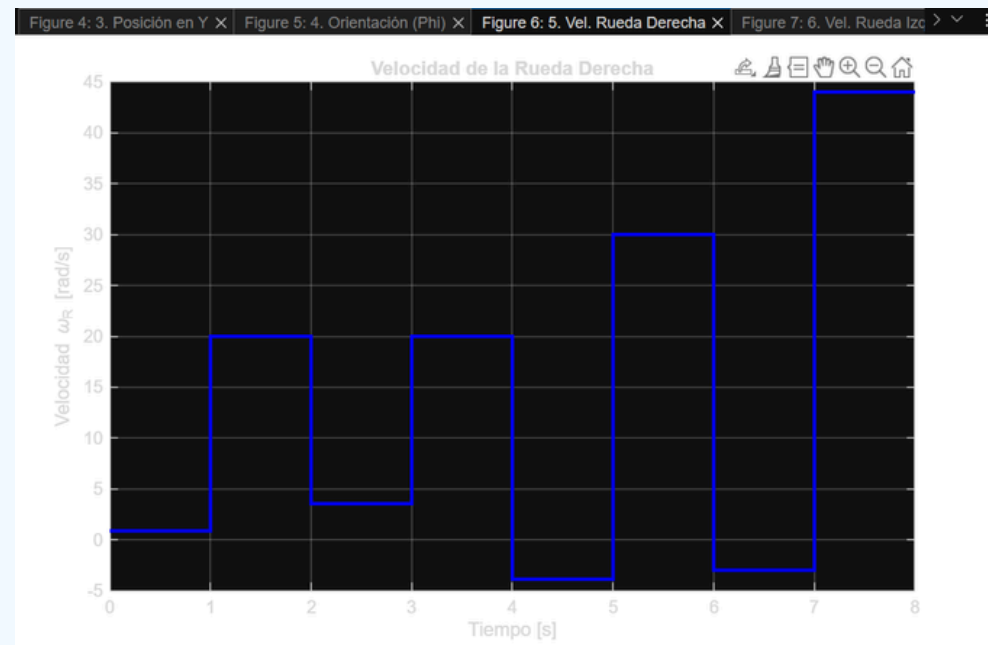
Posición y



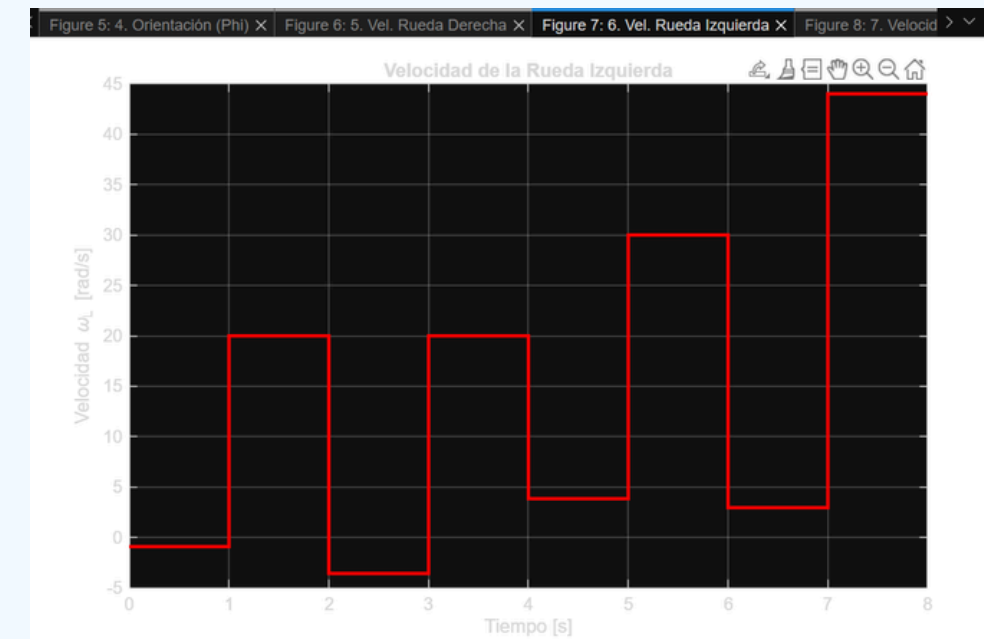
Rotación



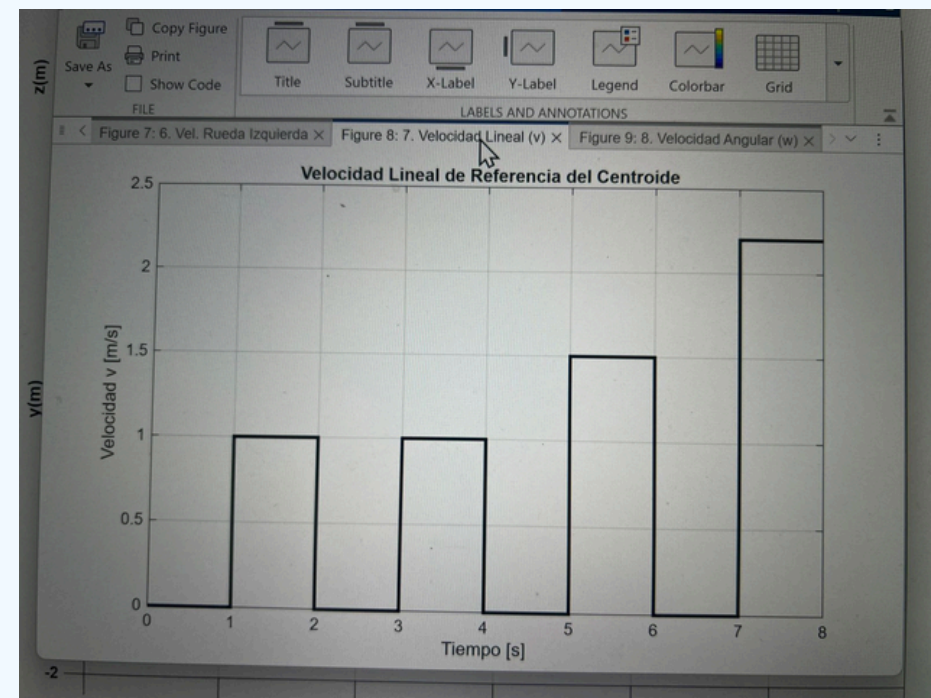
Analisis de graficas



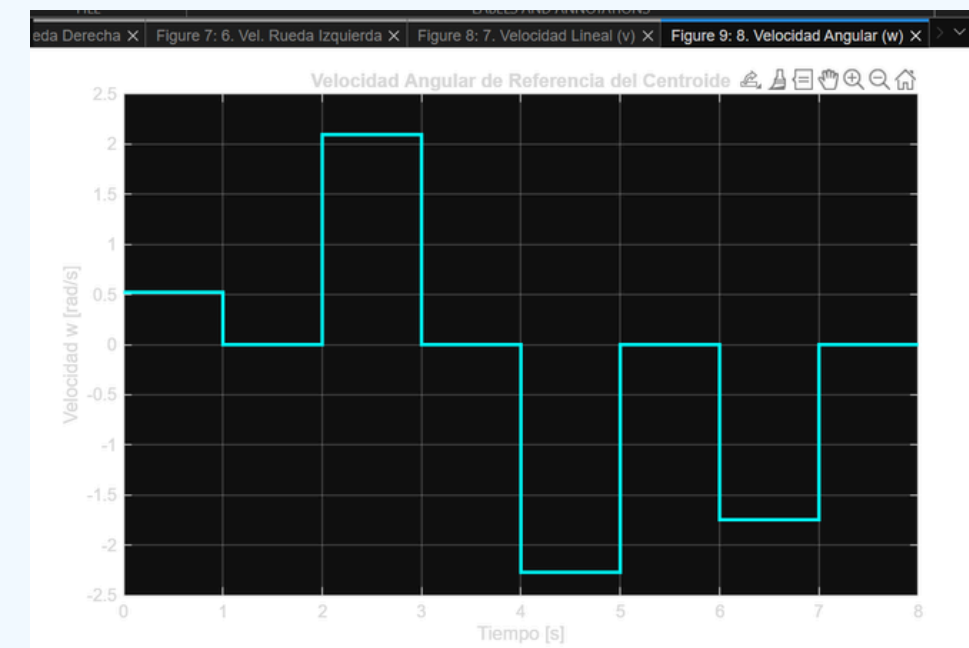
Velocida de la rueda derecha



Velocida de la rueda izquierda



Velocidad lineal



Velocidad angular

Conclusion

Las gráficas de MATLAB representan un escenario ideal que no considera factores físicos como la zona muerta [0.05 m/s] o la saturación [0.4 m/s] propios del hardware real.

